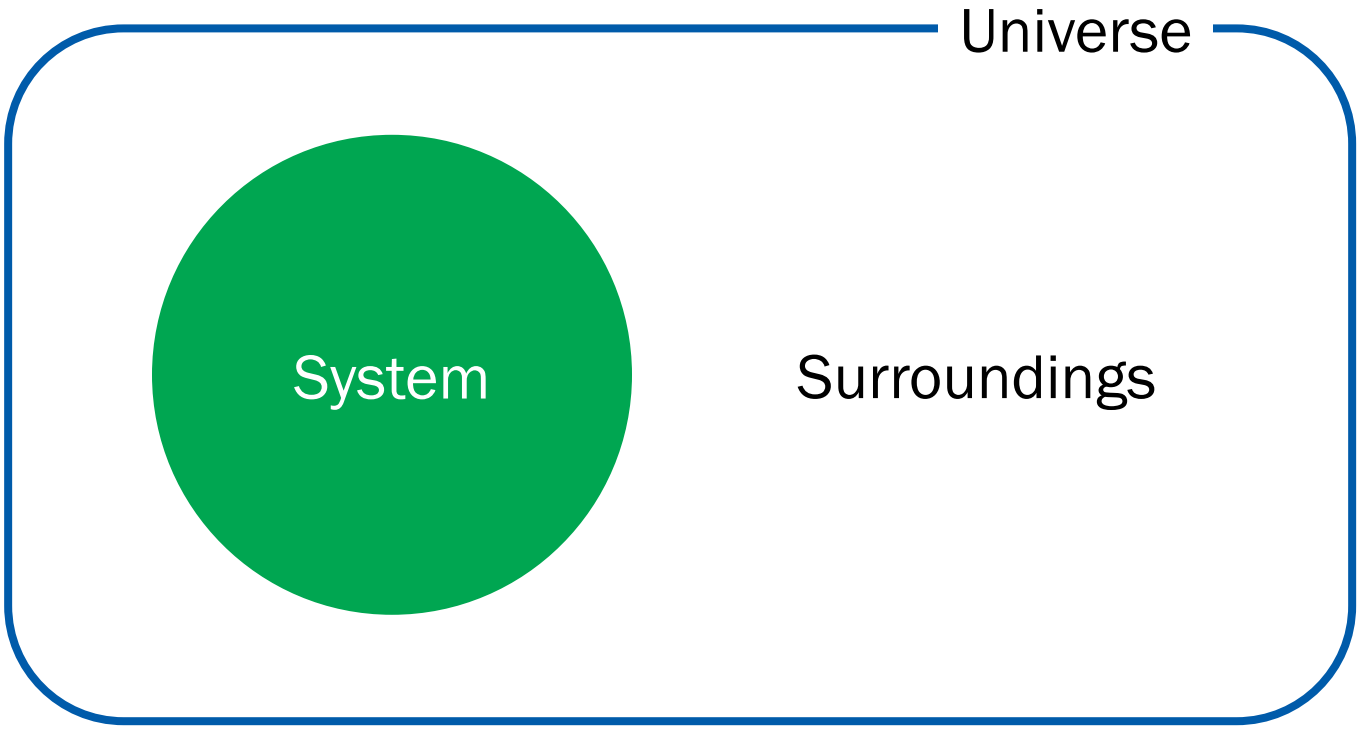
Physical Chemistry of Materials

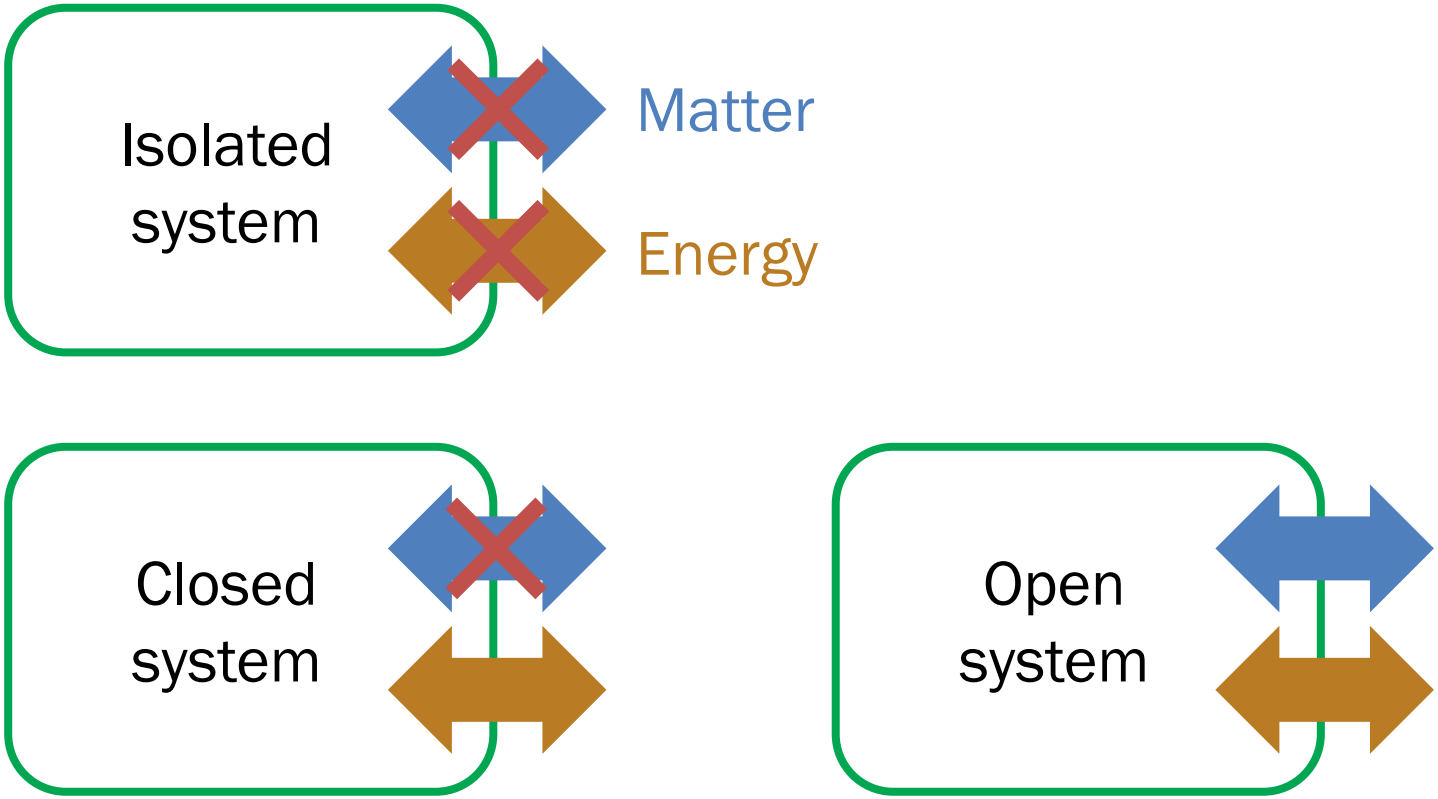
# Lecture 02

## Basic Thermodynamics

# System and Surroundings



# Classification of Systems



# System Properties

- A system can have different physical properties
- A single value will be assigned for each property

- Useful classification
  - Extensive properties: $f(\mathrm{A}) + f(\mathrm{B}) = f(\mathrm{A} + \mathrm{B})$
  - Intensive properties: $f(\mathrm{A}) + f(\mathrm{B}) \neq f(\mathrm{A} + \mathrm{B})$

# Some Extensive Properties

- Mass $m$: measure of the quantity of matter present (unit: kilogram, kg)
- Volume $V$: measure of the quantity of space the sample occupies (unit: cubic meter, $m^3$)
- Amount of substance $n$: measure of the number of specified entities (unit: mole, mol)

# Some Intensive Properties

- Pressure $p$: force $F$ / area $A$ (unit: pascal, Pa)
  - Standard pressure $p^{\ominus}$ = $10^5$ Pa = 1 bar
  - Atmospheric pressure 101325 Pa = 1 atm

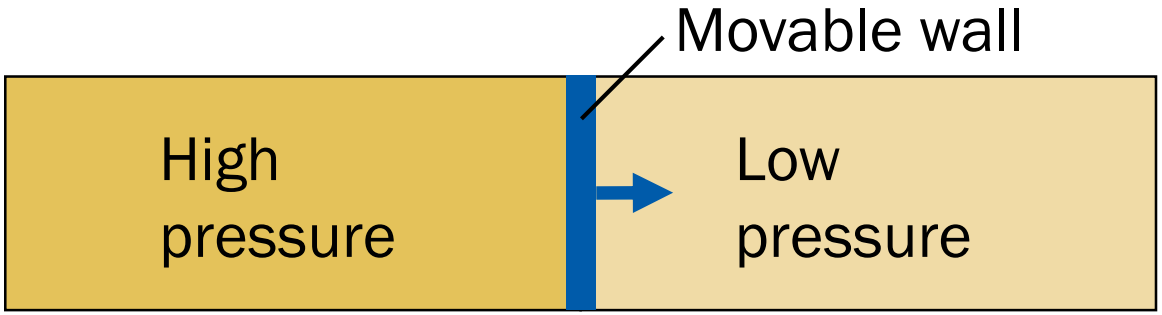
- Temperature $T$: determines the direction of energy flow (unit: kelvin, K)
  - Celsius scale $\theta/^{\circ}\mathrm{C} = T/\mathrm{K} - 273.15$

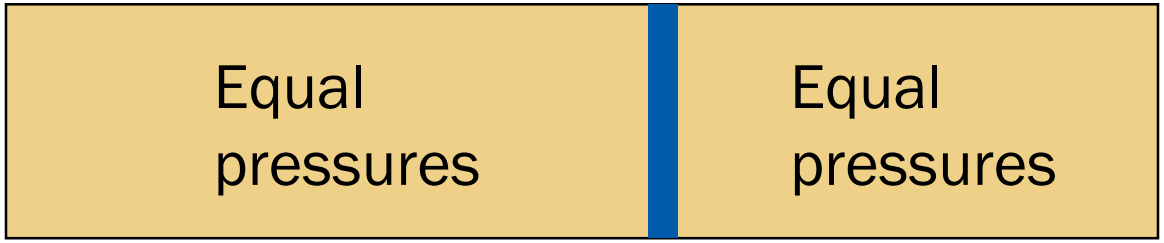
# Equilibrium

- An isolated system is in equilibrium when its properties remain constant with time

# Mechanical Equilibrium

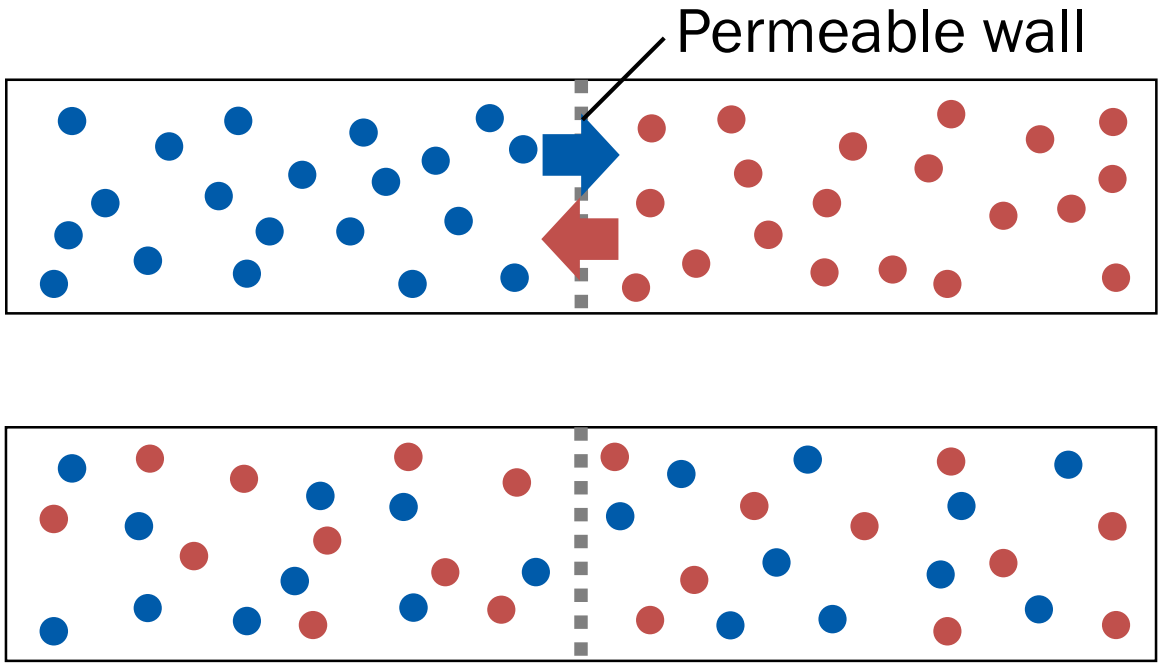
Constant $p$ and $V$



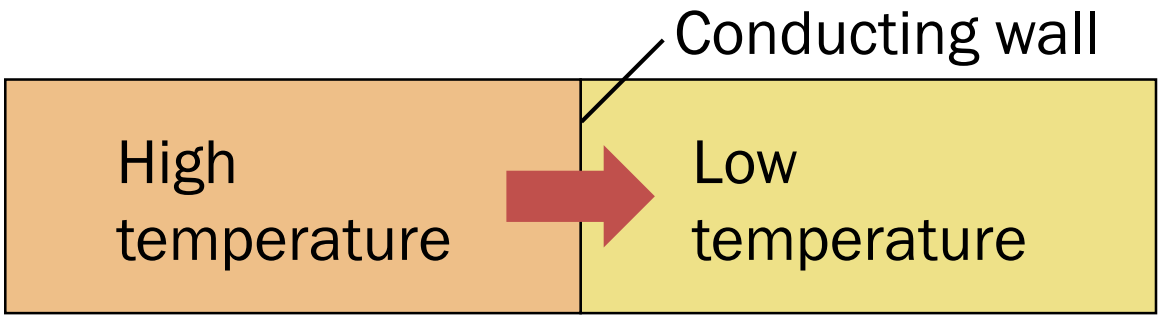
# Diffusive Equilibrium

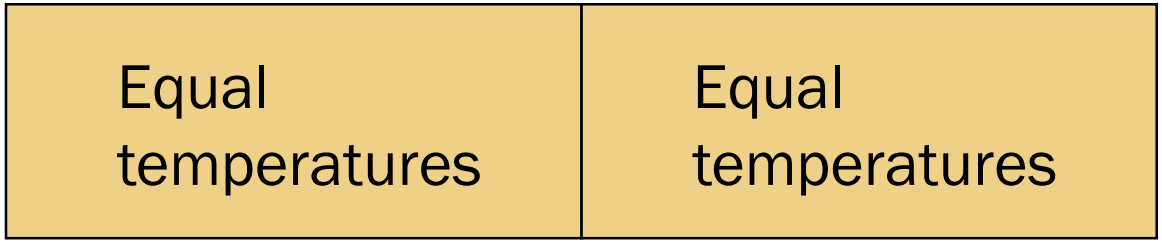
Constant $n$



# Thermal Equilibrium

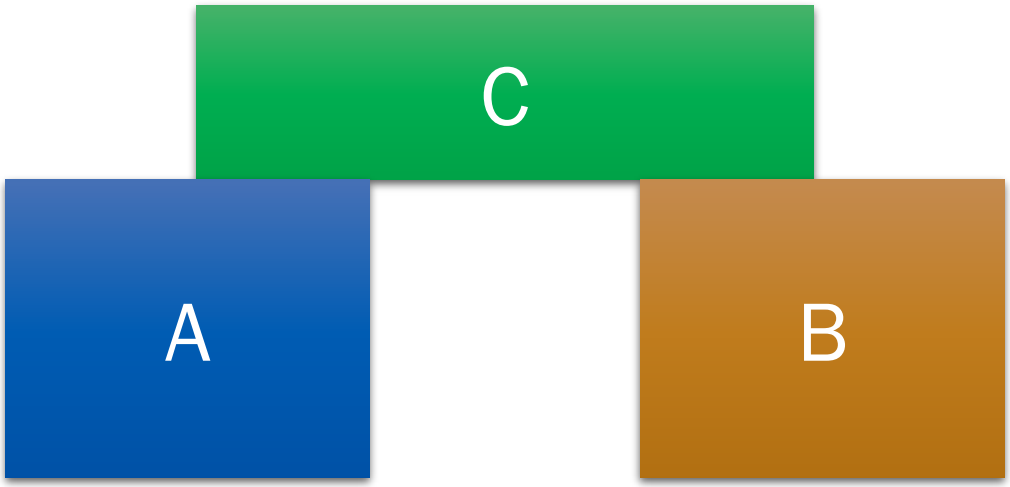
Constant $T$



# Zeroth Law of Thermodynamics

"If systems A and C are in thermal equilibrium, and systems C and B are in thermal equilibrium, systems A and B are in thermal equilibrium."



# State of a System

- The state of a system is defined by its properties
- Two samples of the same substance that have the same properties are in the same state
- Pure substance: $(n, V, p, T)$

# State Change (Process)

The system can change its state

from one to another

$$(n_1, V_1, p_1, T_1) \rightarrow (n_2, V_2, p_2, T_2)$$

# Equation of State

- The state is defined by $(n, V, p, T)$
- Equation of state: relationship of the 4 variables

$$p = f(T, V, n)$$

- Depends on a system (*e.g.* ideal gas)

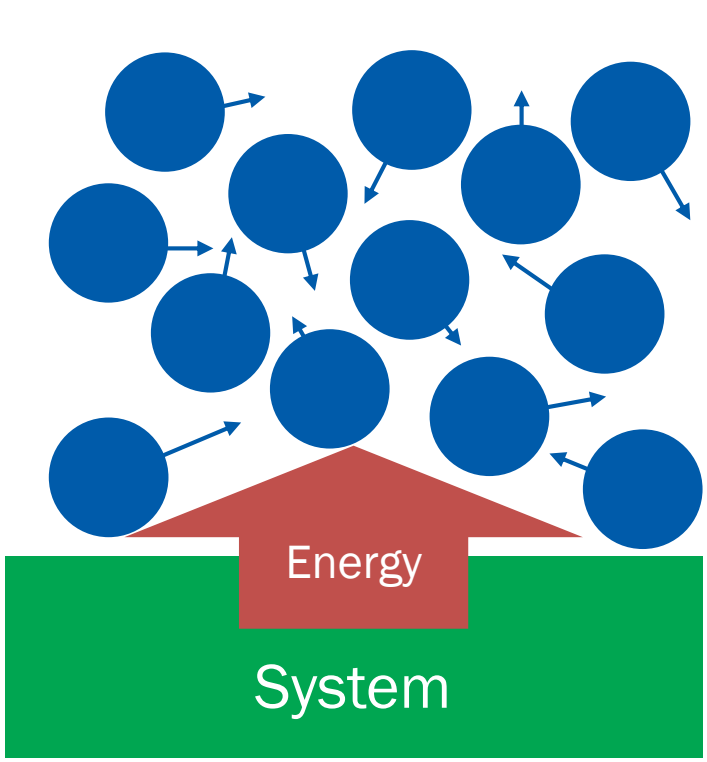
# Internal Energy, $U$

"The total energy of a system"

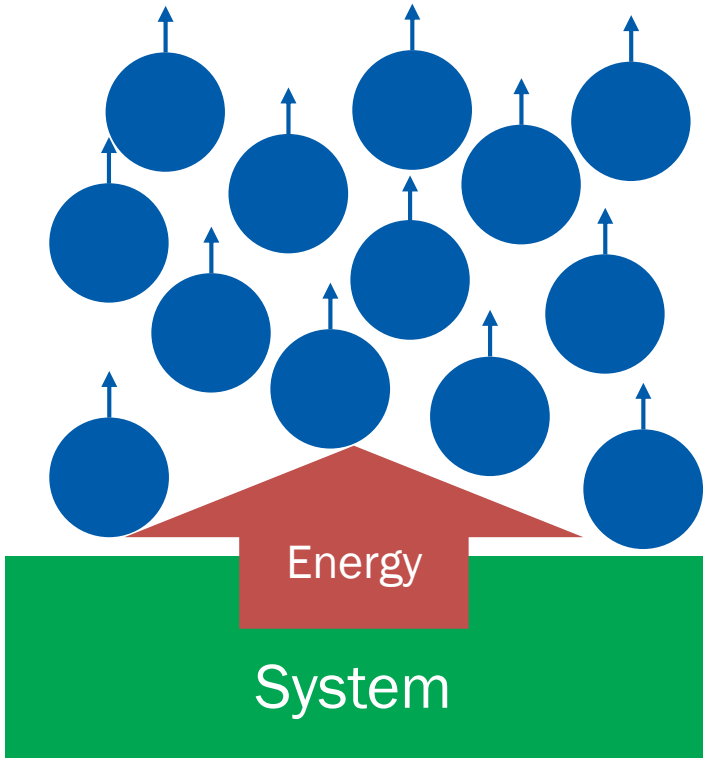
- Potential energy + kinetic energy of all particles
- Almost impossible to obtain absolute values

How can we change the internal energy?

# Two Possible Ways




“Heat”




“Work”

# First Law of Thermodynamics

"The internal energy of an isolated system is constant."

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$\mathrm{d}U = \delta q + \delta w$$

small change of $U$ | small $q$ | small $w$

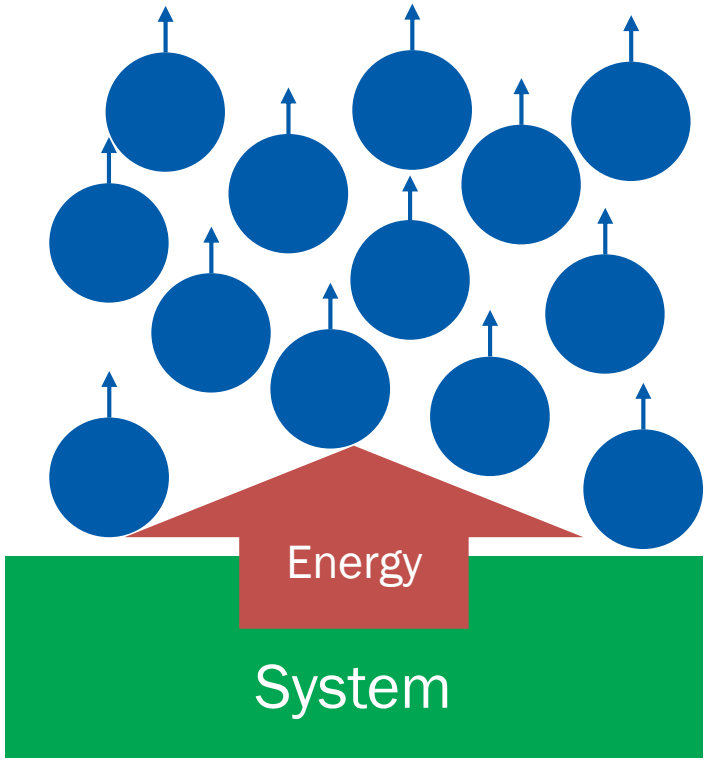
# Work

$$\delta w = \vec{F} \cdot \mathrm{d}\vec{s}$$

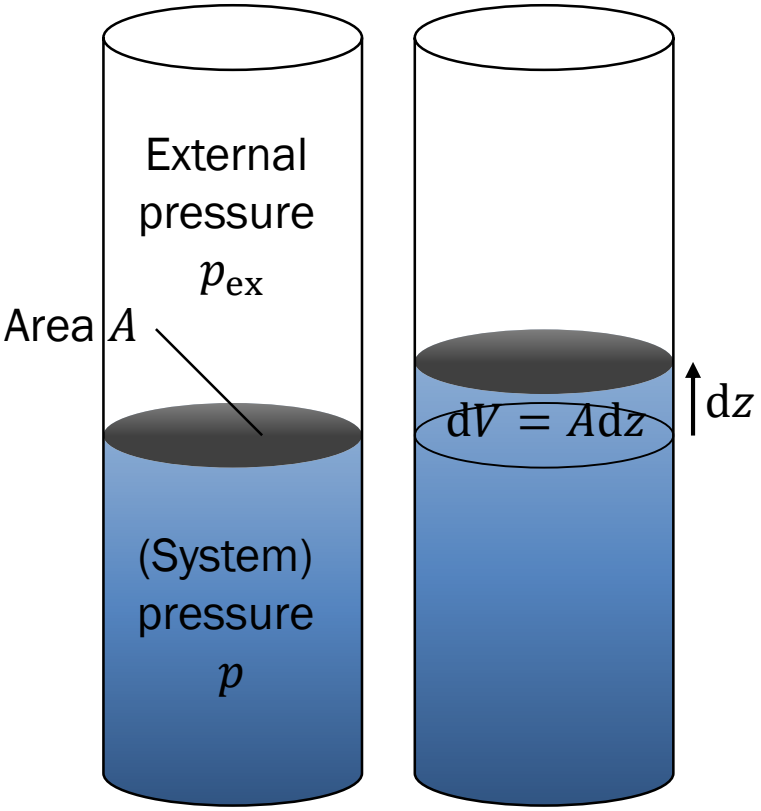
$\vec{F}$: "force" (intensive)

$\mathrm{d}\vec{s}$: "displacement" (extensive)

- Sign indicates the direction of internal energy change



# Expansion Work




$$\delta w = \vec{F} \cdot d\vec{s} = F_x dx + F_y dy + F_z dz$$

As $dx = dy = 0$,

$$\delta w = -|F| dz$$

Since $p_{ex} = |F|/A$,

$$\delta w = -p_{ex} A dz = -p_{ex} dV$$

# State and Path Functions

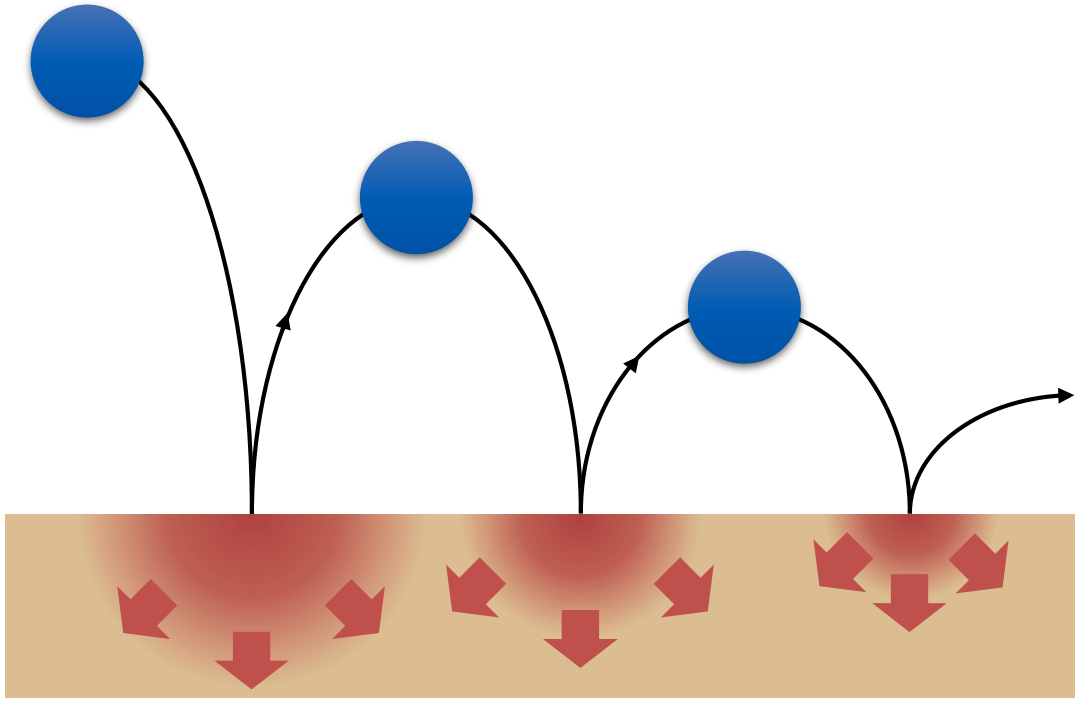
- State function: a property that depends only on the current state of a system and is independent of its history

  *e.g.* internal energy, pressure, volume, temperature

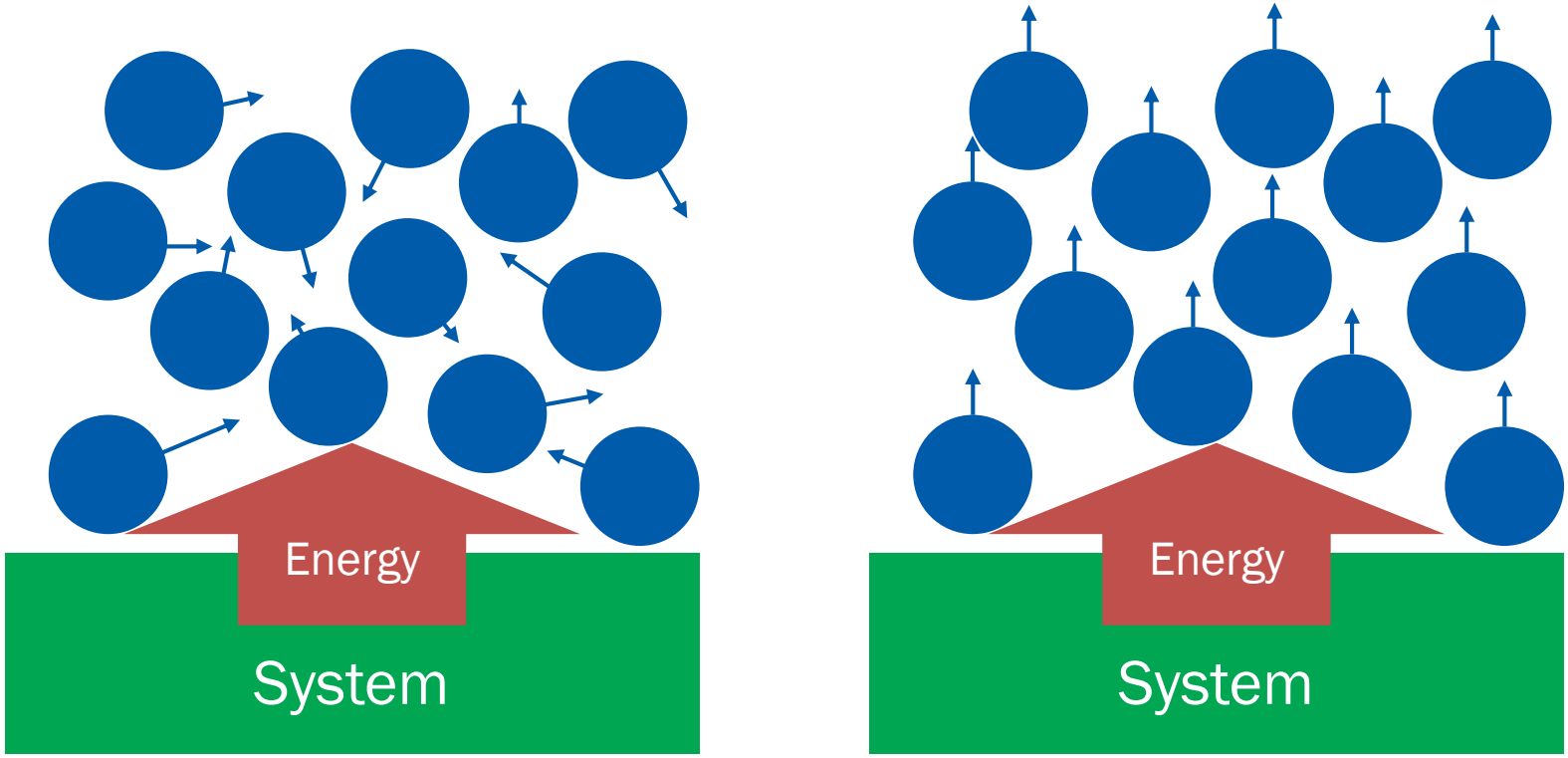
- Path function: a property with a value that depends on the path between two states

  *e.g.* heat, work

# “Arrow of Time”



# Why?




Heat and work are not equal!

# Reversible vs. Irreversible Changes

- Reversible change consists of infinitesimal changes where the system is always in equilibrium
  - The reverse makes perfect sense
- Irreversible change involves non-equilibrium changes
  - The reverse looks strange
  - Arrow of time!

# Entropy

$$\mathrm{d}S = \frac{\delta q_{\mathrm{rev}}}{T}$$

- Index of exhaustion
- Second law of thermodynamics

$$\mathrm{d}S \geq 0 \text{ (isolated system)}$$

  - $\mathrm{d}S = 0$: reversible change
  - $\mathrm{d}S > 0$: irreversible change

# Third Law of Thermodynamics

The entropy of all perfect crystalline substances

is zero at $T = 0$

$$\lim_{T \to 0} S = 0$$

# Clausius Inequality

$$\mathrm{d}S_{\text{universe}} = \mathrm{d}S_{\text{sys}} + \mathrm{d}S_{\text{surr}} \geq 0$$

For the surroundings, every change is reversible;

$$\mathrm{d}S_{\text{surr}} = \frac{\delta q_{\text{surr}}}{T}$$

Since $\delta q_{\text{surr}} = -\delta q_{\text{sys}}$,

$$\mathrm{d}S_{\text{sys}} - \frac{\delta q_{\text{sys}}}{T} \geq 0$$

$$\therefore \mathrm{d}S \geq \frac{\delta q}{T}$$

# Fundamental Equation of Thermodynamics

From the first law,

$$\mathrm{d}U = \delta q + \delta w$$

For a reversible change,

$$\mathrm{d}U = \delta q_{\mathrm{rev}} + \delta w_{\mathrm{rev}} = T\mathrm{d}S - p\mathrm{d}V$$

Since all variables are state functions,

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V \text{ (in general)}$$

# Natural Independent Variables

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

$\rightarrow S$ and $V$ are natural independent variables of $U$

$$U = U(S, V)$$

As $U$ is a state function,

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_V \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_S \mathrm{d}V$$

# Thermodynamic Definitions

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

As $U = U(S, V)$ is a state function,

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_V \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_S \mathrm{d}V$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S}\right)_V$$

$$p = -\left(\frac{\partial U}{\partial V}\right)_S$$

# Euler's Theorem

If $x = x(y, z)$ for extensive properties $x$, $y$, and $z$,

$$x = \left(\frac{\partial x}{\partial y}\right)_z y + \left(\frac{\partial x}{\partial z}\right)_y z$$

Hence, for $U = U(S, V)$,

$$U = \left(\frac{\partial U}{\partial S}\right)_V S + \left(\frac{\partial U}{\partial V}\right)_S V = TS - pV$$

conjugate pairs

# Legendre Transforms (1)

$$U = U(S, V)$$

Can we get another function with different variables?

Subtract a conjugate pair!

- $(S, V) \rightarrow (S, p)$

$$H = U + pV \text{ (enthalpy)}$$

$$\mathrm{d}H = \mathrm{d}U + \mathrm{d}(pV) = T\mathrm{d}S + V\mathrm{d}p$$

$$\therefore H = H(S, p)$$

# Legendre Transforms (2)

$$U = U(S, V)$$

Can we get another function with different variables?

Subtract a conjugate pair!

- $(S, V) \rightarrow (T, V)$

$$A = U - TS \text{ (Helmholtz energy)}$$

$$\mathrm{d}A = \mathrm{d}U - \mathrm{d}(TS) = -S\mathrm{d}T - p\mathrm{d}V$$

$$\therefore A = A(T, V)$$

# Legendre Transforms (3)

$$U = U(S, V)$$

Can we get another function with different variables?

Subtract a conjugate pair!

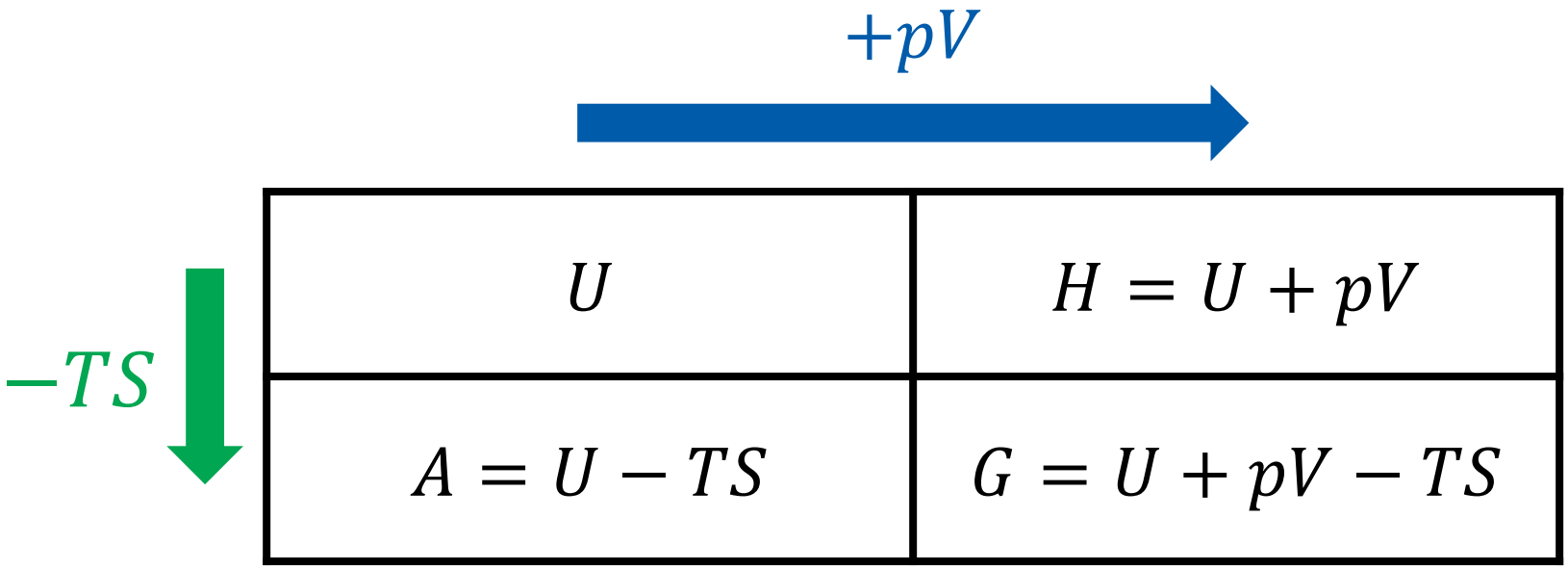
- $(S, V) \rightarrow (T, p)$

$$G = U + pV - TS \text{ (Gibbs energy)}$$

$$\mathrm{d}G = \mathrm{d}U + \mathrm{d}(pV) - \mathrm{d}(TS) = -S\mathrm{d}T + V\mathrm{d}p$$

$$\therefore G = G(T, p)$$

# Thermodynamic Potentials



# Natural Independent Variables

- Internal energy $U = U(S, V)$

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Enthalpy $H = U + pV = H(S, p)$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$$

# Two Systems in Equilibrium

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$$

- Mechanistic equilibrium: $p_1 = p_2$
  - Conjugate extensive variable: $\mathrm{d}V$
- Thermal equilibrium: $T_1 = T_2$
  - Conjugate extensive variable: $\mathrm{d}S = \delta q/T$
- Diffusive equilibrium: ?
  - Conjugate extensive variable: $\mathrm{d}n$

# Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
  - Conjugate extensive variable: $\mathrm{d}n$

- (Modified) differentials of thermodynamic potentials

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + \mu\mathrm{d}n$$

$$\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + \mu\mathrm{d}n$$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

# Three Independent Variables

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

This implies that $U = U(S, V, n)$, which leads to

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_{V,n} \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_{S,n} \mathrm{d}V + \left(\frac{\partial U}{\partial n}\right)_{S,V} \mathrm{d}n$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \qquad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \qquad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

# Meaning of $\mu$

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

This implies that $G = G(T, p, n)$.

$G$ is extensive, and $n$ is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T,p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

# Meaning of $\mu$

$$g(T,p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n}\right)_{T,p}$$

However,

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu\mathrm{d}n$$

gives

$$\mu = \left(\frac{\partial G}{\partial n}\right)_{T,p} = g(T,p) = \frac{G}{n}$$

(molar Gibbs energy)

# Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$

# Two Systems in Equilibrium

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu\mathrm{d}n$$

- Mechanistic equilibrium: $p_1 = p_2$
  - Conjugate extensive variable: $\mathrm{d}V$
- Thermal equilibrium: $T_1 = T_2$
  - Conjugate extensive variable: $\mathrm{d}S = \delta q/T$
- Diffusive equilibrium: $\mu_1 = \mu_2$
  - Conjugate extensive variable: $\mathrm{d}n$

# Multi-Component System

- State: $(p, T, n) \rightarrow (p, T, n_{\mathrm{A}}, n_{\mathrm{B}}, \cdots)$

- Thermodynamic potentials
  - $\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$
  - $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$
  - $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$
  - $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \mu_{\mathrm{A}}\mathrm{d}n_{\mathrm{A}} + \mu_{\mathrm{B}}\mathrm{d}n_{\mathrm{B}} + \cdots$

# Thermodynamics of Phases

At equilibrium, the chemical potential of a substance is the same in and throughout every phase in the system

